



Project Proposal

SentinelAI

*AI-Powered Multi-Modal Digital Threat Detection & Digital Authenticity
Analysis Platform*

**Software Project Design and Development
CSE 416**

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Executive Summary & Industry Problem Statement

The Digital Authenticity & Cybersecurity Problem

The rapid advancement of Generative AI has made digital manipulation increasingly realistic. **Deepfake images, videos, voices, manipulated government documents, phishing websites, scam messages, and malicious files** can be used for identity impersonation, fraud, misinformation, and cybercrime.

Deepfake technology presents a particularly serious risk because realistic synthetic media can be used to impersonate individuals, fabricate evidence, manipulate official documents, and support fraudulent activities. Users currently need separate tools to investigate different types of digital threats.

The SentinelAI Software Solution

SentinelAI is a web-based AI-powered platform designed to provide a unified analysis system for suspicious digital content. The platform analyzes:

- Deepfake images, videos, and audio
- Potentially manipulated government/official documents
- Phishing URLs and websites
- Scam and social-engineering messages
- Potentially malicious files

The system combines specialized AI detection models with **risk assessment and local LLM analysis using Llama 3.1 / Qwen3 through Ollama** to generate understandable security reports.

Sectoral & Security Impact

By combining multiple threat-detection capabilities into one platform, SentinelAI reduces the need for users to operate multiple specialized tools. The system provides a unified risk score, evidence, explanation, and recommended action.

Academic Scope & 14-Week Execution Horizon

The project is designed to be executed across a **14-week academic timeline**, culminating in a functional web demonstration, integrated AI detection framework, database, and final academic presentation.

1. Project Objectives

1.1 General Objective

The primary objective of SentinelAI is to design, develop, test, and deploy a secure web-based platform capable of detecting **deepfakes, document manipulation, phishing, scams, and malicious files** through AI-assisted analysis.

1.2 Specific Measurable Objectives

1. **Deepfake Detection:** Detect potentially AI-generated or manipulated images, videos, and audio.
2. **Document Manipulation Detection:** Identify technical indicators of possible manipulation in digital and official documents.
3. **Phishing Detection:** Analyze URLs and website characteristics to identify potential phishing threats.
4. **Scam Detection:** Analyze messages and emails for scam and social-engineering patterns.
5. **Malicious File Detection:** Perform static analysis of uploaded files to identify suspicious characteristics.
6. **Risk Assessment:** Combine detection results into a unified risk score and threat classification.
7. **Local LLM Analysis:** Use Ollama with **Llama 3.1 / Qwen3** to explain findings and generate recommendations.
8. **Security Reporting:** Generate a structured report containing the verdict, evidence, risk level, and recommended action.

2. Project Scope & System Boundary

2.1 In-Scope (Minimum Viable Product — MVP)

User Authentication & Profile Subsystem: Secure registration, login authentication, and profile management.

Deepfake Detection Module: Analysis of images, videos, and audio for potential AI generation or manipulation.

Document Analysis Module: Analysis of document images and files for possible manipulation indicators such as visual inconsistencies and metadata anomalies.

Phishing Detection Module: URL and webpage analysis for potentially malicious or phishing content.

Scam Detection Module: NLP-based analysis of suspicious emails, SMS, and messages.

Malicious File Analysis Module: Static analysis of supported executable and document files without executing them.

Risk Assessment Engine: Aggregates detection results and produces a unified threat score.

Local LLM Analysis: Uses Ollama with Llama 3.1 / Qwen3 to interpret findings and generate explanations.

Security Report System: Provides analysis results, evidence, recommendations, and analysis history.

2.2 Out-of-Scope (Explicit Exclusions)

- Automatic legal certification of government documents.
- Automatically declaring a document legally forged.
- Live execution of uploaded malware.
- Autonomous malware removal.
- Full enterprise SOC functionality.
- Training a general-purpose LLM from scratch.

2.3 Target User Roles & Responsibilities Matrix

User: Submits suspicious content, views analysis results, and downloads reports.

Analyst: Reviews suspicious analyses and examines detected indicators.

Administrator: Manages users, monitors analyses, and manages system operations.

3. Comprehensive Feasibility Study

3.1 Technical Feasibility

The proposed technology stack consists of established open-source technologies including **Next.js**, **FastAPI**, **Python**, **PyTorch**, **Scikit-learn**, **OpenCV**, **PostgreSQL**, **Ollama**, **Llama 3.1**, and **Qwen3**.

The modular architecture allows each detection engine to be developed and tested independently before integration.

3.2 Economic Feasibility

The academic prototype will primarily use open-source software, local AI models, and free development infrastructure.

Estimated Academic Cost: Approximately 30,000 BDT, primarily for domain and basic deployment requirements.

3.3 Operational Feasibility

SentinelAI replaces a fragmented investigation process with a simple workflow:

Submit → Analyze → Risk Assessment → Explanation → Report

3.4 Schedule Feasibility (14-Week Execution Horizon)

The project will follow a structured 14-week development schedule covering requirements, AI development, backend, frontend, integration, testing, and deployment.

4. Work Breakdown Structure (WBS)

4.1 Deep Hierarchical WBS Task Breakdown

- **Project Initiation & Architecture**
 - Define threat categories, requirements, and system boundaries.
 - Design system architecture and database ERD.
- **AI/ML Model Development**
 - Prepare deepfake, document, phishing, scam, and malware datasets.
 - Develop and evaluate specialized detection models.
- **Backend & Database Development**
 - Initialize FastAPI backend, PostgreSQL database, and authentication.
 - Develop analysis APIs and unified risk-scoring engine.
- **Local LLM Integration**
 - Configure Ollama and integrate Llama 3.1 / Qwen3.
 - Develop evidence analysis, explanation, and recommendation generation.
- **Frontend Client Application Development**
 - Initialize Next.js application and responsive UI.
 - Build analysis forms, dashboards, result pages, and reports.
- **Testing, QA & Deployment**
 - Execute model, API, security, and integration testing.
 - Deploy the application and database to the cloud environment.

4.2 WBS Responsibility Matrix

WBS ID	Work Package	Summary Deliverable	Responsible
1.0	Project Initiation	Requirements, ERD, Architecture	All Members
2.0	AI/ML Development	Detection Models	AI/ML Members
3.0	Backend Development	APIs, Database, Risk Engine	Backend Member
4.0	LLM Integration	Ollama + Llama 3.1/Qwen3	AI/Backend Members
5.0	Frontend	Web UI & Dashboard	Frontend Member
6.0	Testing & Deployment	Tested & Deployed System	All Members

5. Gantt Chart & 14-Week Project Timeline

5.1 14-Week Project Schedule & Dependency Matrix

Task	Start	End	Duration	Dependencies
1. Requirements & Architecture	Week 1	Week 2	2 Weeks	None
2. Dataset & AI Development	Week 2	Week 7	6 Weeks	Task 1
3. Backend & Database	Week 4	Week 8	5 Weeks	Task 1
4. LLM Integration	Week 7	Week 9	3 Weeks	Task 2, 3
5. Frontend Development	Week 6	Week 10	5 Weeks	Task 1
6. System Integration	Week 10	Week 12	3 Weeks	Task 3, 4, 5
7. Security & Performance Testing	Week 12	Week 13	2 Weeks	Task 6
8. Deployment & Presentation	Week 13	Week 14	2 Weeks	Task 7

6. Budget & Financial Plan

6.1 Estimated Resource Budget

Item	Specification	Estimated Cost
Domain	.com / .xyz, 1 Year	~2,000 BDT
Cloud Hosting	Production-grade server	~15,000 BDT/month
PostgreSQL	Managed production database	~8,000 BDT/month
Ollama	Dedicated local LLM server infrastructure	~5,000 BDT/month
Llama 3.1 / Qwen3	Self-hosted local models	0 BDT
ML Frameworks	PyTorch / Scikit-learn / OpenCV	0 BDT
GitHub	Professional repository & CI/CD	~2,000 BDT/month
SSL	Commercial/managed SSL	~3,000 BDT/year

6.2 Financial Summary

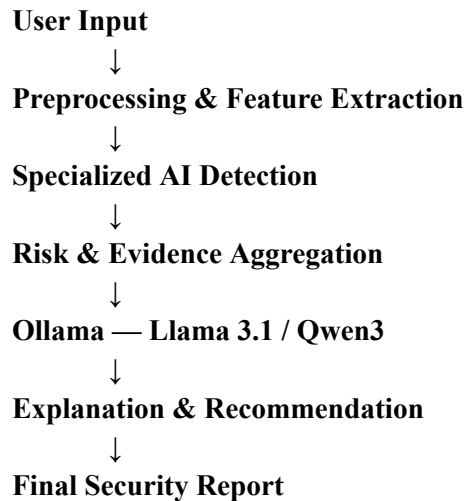
Estimated Academic Project Cost: Approximately 30,000 BDT.

The project minimizes operational expenses by using open-source AI models, local LLM inference, and free development infrastructure.

7. System Specifications & Technical Architecture

7.1 Proposed Processing Architecture

The system follows a simple unified methodology:



The specialized detection engines handle deepfake, document, phishing, scam, and malware analysis, while the local LLM provides contextual interpretation of the generated findings.

7.2 Core Technology Stack

- **Frontend:** Next.js, TypeScript
- **Backend:** FastAPI, Python
- **AI/ML:** PyTorch, Scikit-learn, OpenCV
- **Local LLM:** Ollama, Llama 3.1, Qwen3
- **Database:** PostgreSQL
- **Deployment:** Docker, Render

7.3 Security Hardening & Data Protection Controls

The system will implement secure authentication, password hashing, input validation, file-type restrictions, secure file handling, parameterized database queries, rate limiting, and audit logging.

Uploaded malicious files will be treated as **untrusted input** and analyzed statically rather than executed directly on the production server.

8. Related Works & Industry Competitors

Platform	Primary Focus
VirusTotal	Malware & URL Analysis
Google Safe Browsing	Phishing & Malicious Websites
Microsoft Defender SmartScreen	Web & File Protection
Reality Defender	Deepfake & Synthetic Media
SentinelAI	Deepfake + Document + Phishing + Scam + Malware

Proposed Gap

Existing solutions generally specialize in individual threat categories. **SentinelAI proposes a unified web-based platform combining digital authenticity analysis and cybersecurity threat detection, with local LLM-based explanation and unified risk assessment.**

9. References & Academic Citations

1. VirusTotal. VirusTotal Documentation & API Reference.
2. Google. Google Safe Browsing Documentation.
3. Microsoft. Microsoft Defender SmartScreen Documentation.
4. Reality Defender. AI-Generated Media Detection Documentation.
5. PyTorch. PyTorch Documentation.
6. Scikit-learn. Machine Learning Documentation.
7. OpenCV. Open Source Computer Vision Library Documentation.
8. Ollama. Ollama Documentation.
9. Meta AI. Llama 3.1 Documentation.
10. Qwen Team. Qwen3 Documentation.
11. PostgreSQL Global Development Group. PostgreSQL Documentation.
12. FastAPI. FastAPI Documentation.